BEMT Project

Aircraft propeller design and analysis

AERO0014

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1 Problem statement

With recent advances in electric propulsion, power electronics and energy storage, a new class of aircraft relying on Distributed Electric Propulsion (DEP) has emerged. Instead of a small number of highly loaded propellers, DEP aircraft use several moderately loaded propellers distributed along the airframe in order to improve takeoff performance.

In this context, you are asked to design and analyze one propeller of a distributed electric propulsion system. This propeller is driven by an independent electric motor operating at constant rotational speed.

Figure 1: Distributed electric propulsion STOL demonstrator aircraft (NASA X-57 Maxwell)[1].

This project will therefore consist, firstly, in the design of the propeller, using the Adkins and Liebeck[2] methodology in takeoff conditions. The main geometry parameters used as an input of this design algorithm are summarized in Table 1.

Table 1: Main propeller characteristics

Parameter	**Value**
Number of blades	3
Airfoil	Clark-Y
Tip diameter	1.7 m
Hub diameter	0.3 m

In the second part of the project, you will have to implement the Blade Element Momentum Theory algorithm to analyze your design, discuss the physics behind the propeller, and properly choose the control strategy to allow for efficient cruise operation.

As specified in the Table 1, this propeller uses a Clark-Y airfoil along the entire blade. Python scripts are thus provided in order to interpolate $C_{\mathcal{L}}$ and $C_{\mathcal{D}}$ values directly from the polars of this airfoil.

For the project, you are asked to:

1. Perform the propeller design at takeoff conditions, considering a required thrust of 500 N, a rotational speed of 2100 RPM, and a free-stream velocity of 45 m/s. The propeller must be designed such that each blade section operates at its optimal aerodynamic point, corresponding to the minimum drag-to-lift ratio (c_d/c_l). Present the resulting spanwise distributions of chord and pitch angle, and discuss the main trends observed in the evolution of these geometrical parameters along the blade span. **(7/20)**

2. Implement the bemt algorithm and compute the thrust produced by the propeller, as well as the power it consumes and its efficiency. Plot the thrust, power and efficiency distribution along the blade span. Repeat this process by applying a collective pitch angle of 10°. Compare the results obtained with the two configurations and discuss in terms of velocity triangles. **(6/20)**

3. Plot the thrust and power coefficients, and the propulsive efficiency with respect to the advance ratio for collective pitches of 0, 10, 20 and 30°. Compare the different curves you obtain. **(4/20)**

4. Now, consider cruise conditions, for which you keep the same rotational speed as for takeoff for the propeller, but the velocity of the aircraft becomes 90 m/s. Which strategy would you adopt to achieve the highest possible propulsive efficiency? You can neglect the changes in density and viscosity, and keep the sea level air properties. **(3/20)**

Hints

- The geometry of the propeller is obtained for some *reference stations*. You must use that as a basis for an interpolation in order to define the blade elements for the analysis part of the project.
- Superimpose the curves for the collective pitch angles on the same figure when you plot the performance coefficients and the propulsive efficiency with respect to the advance ratio.

2 Practical details

- The project has to be performed by groups of **2 students**.
- The project will count for 30% of the overall grade for this course.
- The deadline for this project is **16 May 2026 at 23h59**.
- The submission will be on eCampus. Two files must be submitted:
  1. An archive file (*zip*, *rar*, *7zip* or *tag.gz*) containing all the code files. Do not forget small separate functions you wrote for basic things (converting units, etc).
  2. The pdf report which explains the approach and provides the *commented* results. It is important to send the PDF *alongside* the archive and not *inside* it, as it allow us to directly give feedback on the pdf in eCampus.
- Both files must be send out via **eCampus** > Assignments > BEMT project.
  - *If (and only if) there is any issue with eCampus, you can send it by mail at Jerome.VanDenBerghe@uliege.be.*

Code review

All codes will be reviewed and read in detail, as well as checked for plagiarism. As the report is rather small, please provide detailed comments (in English) in the code so I understand your methodology. Well-made comments in the code help me understand what your tried to do in case of an error.

The codes should all run on my computer with minimal intervention on my side. Make sure all small functions that you have used are provided alongside the main script. If your code is in Python, it is strongly advised that you setup a proper virtual environment for this project (and for all projects in general). You are also **required to add a requirements.txt file** alongside your scripts. This file ensures that I will be able to run your code with the same libraries as you are.

The code quality in itself will not be graded, but please try to make it clean anyway. Look into the codes given alongside this statement for examples of well-formatted codes.

REFERENCES

[1] NASA. *X-57 Maxwell Distributed Electric Propulsion Aircraft*. https://sacd.larc.nasa.gov/asab/asab-projects-2/x57maxwell/. Accessed February 2026.

[2] C. N. Adkins and R. H. Liebeck. "Design of optimum propellers". In: *Journal of Propulsion and Power* 10.5 (1994), pp. 676–682.